

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: df = pd.read_csv('mymoviedb.csv', lineterminator = '\n')
```

```
In [4]: df.sample(4)
```

```
Out[4]:
```

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average
5323	1980-09-01	Flash Gordon	A football player and his mates travel to the ...	20.058	691	6.2
2883	2015-10-15	Momentum	When Alex, an infiltration expert with a secre...	31.084	436	5.7
3256	1999-07-17	Pokémon: The Movie 2000	Satoshi must put his skill to the test when he...	28.430	655	6.6
4111	1951-11-08	Quo Vadis	Set against the back drop of Rome in crisis, G...	24.007	284	7.1

```
In [5]: df.info() # overview like summary of data
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Release_Date          9827 non-null   object
1   Title                 9827 non-null   object
2   Overview              9827 non-null   object
3   Popularity            9827 non-null   float64
4   Vote_Count            9827 non-null   int64
5   Vote_Average          9827 non-null   float64
6   Original_Language     9827 non-null   object
7   Genre                 9827 non-null   object
8   Poster_Url           9827 non-null   object
dtypes: float64(2), int64(1), object(6)
memory usage: 691.1+ KB
```

```
In [6]: df.duplicated().sum()
```

```
Out[6]: np.int64(0)
```

```
In [7]: #exploring genres column  
df['Genre']
```

```
Out[7]: 0      Action, Adventure, Science Fiction  
1              Crime, Mystery, Thriller  
2                      Thriller  
3      Animation, Comedy, Family, Fantasy  
4      Action, Adventure, Thriller, War  
      ...  
9822                      Drama, Crime  
9823                      Horror  
9824      Mystery, Thriller, Horror  
9825      Music, Drama, History  
9826      War, Drama, Science Fiction  
Name: Genre, Length: 9827, dtype: object
```

```
In [8]: df.describe() # gives the stastical summary of a ddata
```

```
Out[8]:
```

	Popularity	Vote_Count	Vote_Average
count	9827.000000	9827.000000	9827.000000
mean	40.326088	1392.805536	6.439534
std	108.873998	2611.206907	1.129759
min	13.354000	0.000000	0.000000
25%	16.128500	146.000000	5.900000
50%	21.199000	444.000000	6.500000
75%	35.191500	1376.000000	7.100000
max	5083.954000	31077.000000	10.000000

```
In [ ]:
```

```
In [9]: df.head()
```

Out[9]:

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Orig
--	--------------	-------	----------	------------	------------	--------------	------

0	2021-12-15	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	
1	2022-03-01	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	
2	2022-02-25	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	
3	2021-11-24	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	
4	2021-12-22	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	

In []: *#changing realese_date row data types to date_time*

```
In [10]: df['Release_Date'] = pd.to_datetime(df['Release_Date'])
df.dtypes
```

```
Out[10]: Release_Date      datetime64[ns]
Title                    object
Overview                 object
Popularity               float64
Vote_Count               int64
Vote_Average             float64
Original_Language        object
Genre                   object
Poster_Url               object
dtype: object
```

```
In [11]: #in realease_date only year is necessary
df['Release_Date'] = df['Release_Date'].dt.year
```

```
In [12]: df.head()
```

Out[12]:	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Orig
0	2021	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	
1	2022	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	
2	2022	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	
3	2021	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	
4	2021	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	

REMOVING CERTAIN ROWS WHICH ARE NOT NECESSARY Overview
Original_Language Poster_Url

```
In [13]: cols = ['Overview','Original_Language','Poster_Url']
```

```
In [14]: df.drop(cols,axis=1,inplace=True)
```

```
In [15]: df.head()
```

Out[15]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	8.1	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	6.3	Thriller
3	2021	Encanto	2402.201	5076	7.7	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	7.0	Action, Adventure, Thriller, War

categorizing Vote_Average column with different names

```
In [16]: def categorize_col(df, col, labels):
edges =[df[col].describe()['min'],
df[col].describe()['25%'],
df[col].describe()['50%'],
df[col].describe()['75%'],
df[col].describe()['max']]
df[col] = pd.cut(df[col],edges,labels = labels,duplicates= 'drop')
return df
```

```
In [17]: labels = ['NOT_POPULAR', 'BELOW_AVERAGE', 'AVERAGE', 'POPULAR',]
categorize_col(df, 'Vote_Average', labels)
```

Out[17]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	POPULAR	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	POPULAR	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	BELOW_AVERAGE	Thriller
3	2021	Encanto	2402.201	5076	POPULAR	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	AVERAGE	Action, Adventure, Thriller, War
...	...	...	...	...	...	...
9822	1973	Badlands	13.357	896	POPULAR	Drama, Crime
9823	2020	Violent Delights	13.356	8	NOT_POPULAR	Horror
9824	2016	The Offering	13.355	94	NOT_POPULAR	Mystery, Thriller, Horror
9825	2021	The United States vs. Billie Holiday	13.354	152	AVERAGE	Music, Drama, History
9826	1984	Threads	13.354	186	POPULAR	War, Drama, Science Fiction

9827 rows × 6 columns

In [18]: `df.head()`

```
Out[18]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	POPULAR	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	POPULAR	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	BELOW_AVERAGE	Thriller
3	2021	Encanto	2402.201	5076	POPULAR	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	AVERAGE	Action, Adventure, Thriller, War

```
In [19]: df['Vote_Average'].unique()
```

```
Out[19]: ['POPULAR', 'BELOW_AVERAGE', 'AVERAGE', 'NOT_POPULAR', NaN]
Categories (4, object): ['NOT_POPULAR' < 'BELOW_AVERAGE' < 'AVERAGE' < 'POPULAR']
```

```
In [20]: df['Vote_Average'].value_counts()
```

```
Out[20]: Vote_Average
NOT_POPULAR    2467
POPULAR        2450
AVERAGE        2412
BELOW_AVERAGE  2398
Name: count, dtype: int64
```

```
In [21]: df.dropna(inplace = True)
df.isna().sum()
```

```
Out[21]: Release_Date    0
Title                  0
Popularity             0
Vote_Count            0
Vote_Average          0
Genre                 0
dtype: int64
```

SPLITTING THE ELEMENTS PRESENT IN GENRE

```
In [24]: df['Genre'] = df['Genre'].str.split(',')
```

```
In [25]: df.head()
```

Out[25]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	POPULAR	[Action, Adventure, Science Fiction]
1	2022	The Batman	3827.658	1151	POPULAR	[Crime, Mystery, Thriller]
2	2022	No Exit	2618.087	122	BELOW_AVERAGE	[Thriller]
3	2021	Encanto	2402.201	5076	POPULAR	[Animation, Comedy, Family, Fantasy]
4	2021	The King's Man	1895.511	1793	AVERAGE	[Action, Adventure, Thriller, War]

In [29]: `#explode the values,
df = df.explode('Genre').reset_index(drop = True)`

In [30]: `df.head()`

Out[30]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	POPULAR	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	POPULAR	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	POPULAR	Science Fiction
3	2022	The Batman	3827.658	1151	POPULAR	Crime
4	2022	The Batman	3827.658	1151	POPULAR	Mystery

In [35]: `#casting columns into category
df['Genre'] = df['Genre'].astype('category')
df['Genre'].dtypes`

```
Out[35]: CategoricalDtype(categories=['Action', 'Adventure', 'Animation', 'Comedy', 'Crime',
                                     'Documentary', 'Drama', 'Family', 'Fantasy', 'History',
                                     'Horror', 'Music', 'Mystery', 'Romance', 'Science Fiction',
                                     'TV Movie', 'Thriller', 'War', 'Western'],
                                     , ordered=False, categories_dtype=object)
```

```
In [36]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25552 entries, 0 to 25551
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   Release_Date    25552 non-null  int32
 1   Title           25552 non-null  object
 2   Popularity      25552 non-null  float64
 3   Vote_Count      25552 non-null  int64
 4   Vote_Average    25552 non-null  category
 5   Genre           25552 non-null  category
dtypes: category(2), float64(1), int32(1), int64(1), object(1)
memory usage: 749.6+ KB
```

```
In [37]: df.nunique()
```

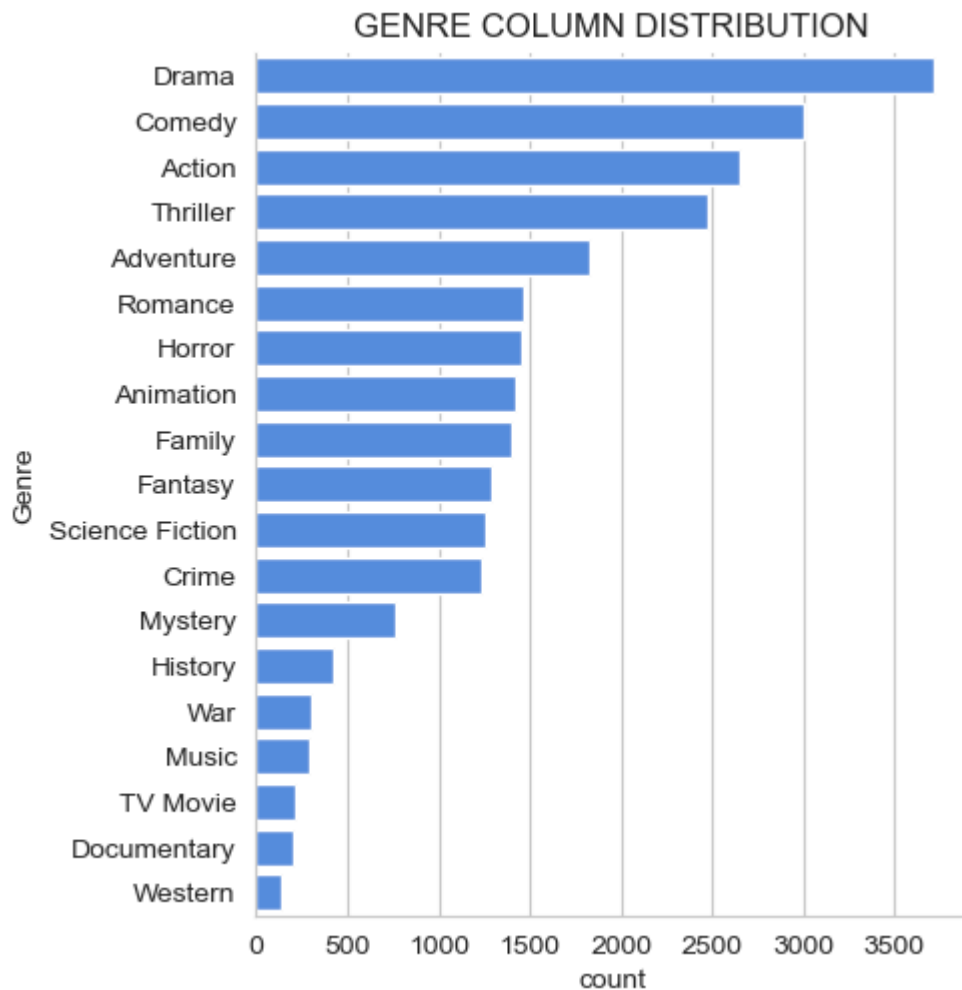
```
Out[37]: Release_Date    100
         Title           9415
         Popularity      8088
         Vote_Count      3265
         Vote_Average     4
         Genre           19
         dtype: int64
```

DATA VISUALIZATION

```
In [38]: sns.set_style('whitegrid')
```

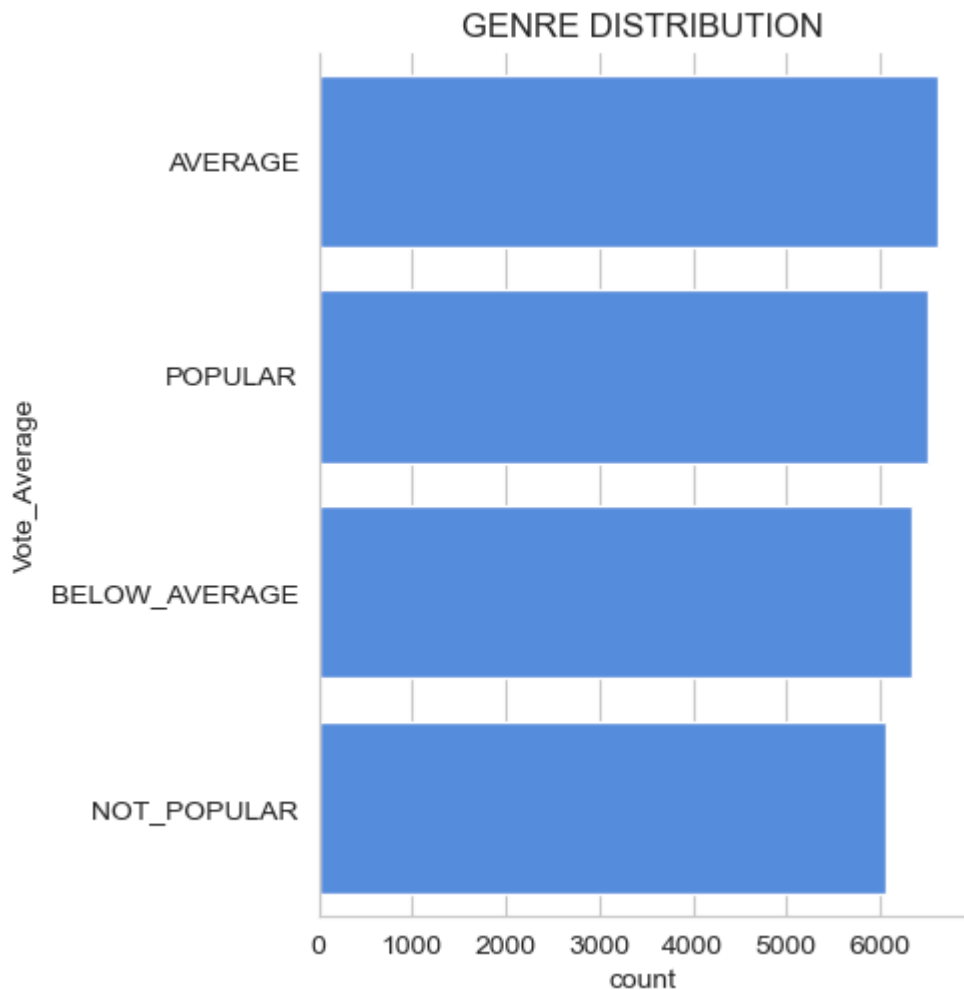
Q1. WHAT IS THE MOST FREQUENT GENRE IN DATASET

```
In [40]: sns.catplot(y = 'Genre', data = df, kind = 'count',
                    order = df['Genre'].value_counts().index,
                    color = '#4287f5')
plt.title('GENRE COLUMN DISTRIBUTION')
plt.show()
```



Q2. WHICH GENRE HAS HIGHEST VOTES

```
In [43]: sns.catplot(y = 'Vote_Average', data = df, kind = 'count',  
                    order = df['Vote_Average'].value_counts().index,  
                    color = '#4287f5')  
plt.title('GENRE DISTRIBUTION')  
plt.show()
```



Q3. WHICH MOVIE GOT HIGHEST POPULARITY ?WHAT IS ITS GENRE?

```
In [45]: df[df['Popularity'] == df['Popularity'].max()]
```

```
Out[45]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	POPULAR	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	POPULAR	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	POPULAR	Science Fiction

Q3. WHICH MOVIE GOT HIGHEST POPULARITY ?WHAT IS ITS GENRE

```
In [51]: df[df['Popularity'] == df['Popularity'].min()].head()
```

```
Out[51]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	13.354	8940	POPULAR	Action
1	2021	Spider-Man: No Way Home	13.354	8940	POPULAR	Adventure
2	2021	Spider-Man: No Way Home	13.354	8940	POPULAR	Science Fiction
3	2022	The Batman	13.354	1151	POPULAR	Crime
4	2022	The Batman	13.354	1151	POPULAR	Mystery

Q5.WHICH YEAR HAS MOST FILMMED MOVIES

```
In [52]: df['Release_Date'].hist()  
plt.title('MOST FILMMED MOVIES YEAR')  
plt.show()
```

